

ReMedX

Project Proposal — Healix AI HealthHack 2026, Track 7 (Drug Discovery & Genomics)

What we're building

ReMedX is a tool for exploring drug repurposing — the idea that a drug already approved for one disease might work for a completely different one, because both conditions are driven by the same underlying gene or protein. Instead of designing a new molecule from scratch, we're helping researchers spot existing, already-safety-tested drugs worth investigating further.

We're approaching Track 7 from two angles: starting from a disease and finding drugs that might be worth investigating, and starting from a compound and getting its actual computed chemical properties. Both are real, working parts of the tool.

Why this problem

Developing a brand-new drug costs somewhere between \$700 million and \$2.6 billion depending on how you count it, and usually takes 10 to 17 years. Most of that time and money goes into things that repurposing can skip entirely, since the drug is already known to be safe in humans. Sildenafil started out as a heart medication before becoming Viagra. A drug once withdrawn for safety reasons is now used to treat leukemia. This already happens — we wanted to make it easier to find these connections in the first place, instead of relying on researchers stumbling onto them through scattered papers and manual literature review.

How it actually works

Part 1 — disease to drug:

- You type in a disease name.
- We query Open Targets, a public research database, to find the genes/proteins actually linked to that disease, each with a real evidence score.
- For each of those, we look for existing drugs that act on the same target — developed for a different disease entirely.
- A locally-run AI model (Ollama) writes a plain-English explanation of the connection, but it's only allowed to use the facts we've actually retrieved. It can't make things up about the mechanism.

Part 2 — compound to properties:

- You enter a compound (as a SMILES string).
- We render its actual structure and calculate real properties using RDKit — molecular weight, LogP, hydrogen bond donors/acceptors, and a Lipinski's Rule of Five check for drug-likeness.

- These numbers come from real chemistry calculations, not a black-box model guessing — we think that's a more honest way to present this than pretending we trained a toxicity predictor overnight.

What's real vs. what we're being careful about

Every score shown — the disease-target evidence score, the compound properties — comes from a real source (Open Targets or RDKit), not something we invented to make the numbers look good. The AI layer only explains facts we've already pulled, so it isn't free to hallucinate a mechanism that sounds convincing but isn't true. Every result is clearly framed as a research hypothesis, not a treatment recommendation — we don't want anyone walking away thinking this diagnoses or prescribes anything.

We're also upfront that this doesn't evaluate drug safety, dosing, or interactions, and the compound module is rule-based chemistry, not a trained ML classifier — we didn't want to overclaim something we didn't actually build in the time we had.

What we built it with

Python and Flask on the backend, a custom HTML/CSS/JS frontend, live queries to the Open Targets GraphQL API, RDKit for the compound module, a locally-run Ollama model for explanations, and SQLite for caching results so repeat searches don't hit the external API again. Everything runs locally right now — deploying it publicly is on our list, not done yet.

How we checked it actually works

- We tested every query against Open Targets' live API by hand before writing it into the code, rather than guessing field names from old tutorials.
- We checked results against known biology — searching Alzheimer's disease correctly surfaces APP and PSEN1, which are genuinely the two most well-established Alzheimer's genes in the literature.
- We checked RDKit's computed properties against known reference values for common drugs like aspirin, to make sure the numbers were actually right.
- We tried both well-studied diseases and rarer ones to make sure the tool fails gracefully (shows an honest "not enough data" message) instead of breaking when data is thin.

Where this could go next

- Deploying it somewhere beyond localhost so it's actually publicly reachable.
- Adding a reverse lookup — starting from a drug and finding diseases it might be worth investigating for, not just the other way around.
- Linking out to actual published literature/citations per candidate, so a researcher has somewhere to go deeper.

Team

Athavur Rahman, Harini T, Abhishek Patel



Healix AI HealthHack 2026 — Track 7: Drug Discovery & Genomics